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Persistent Homology of Music Network with Three Different Distances

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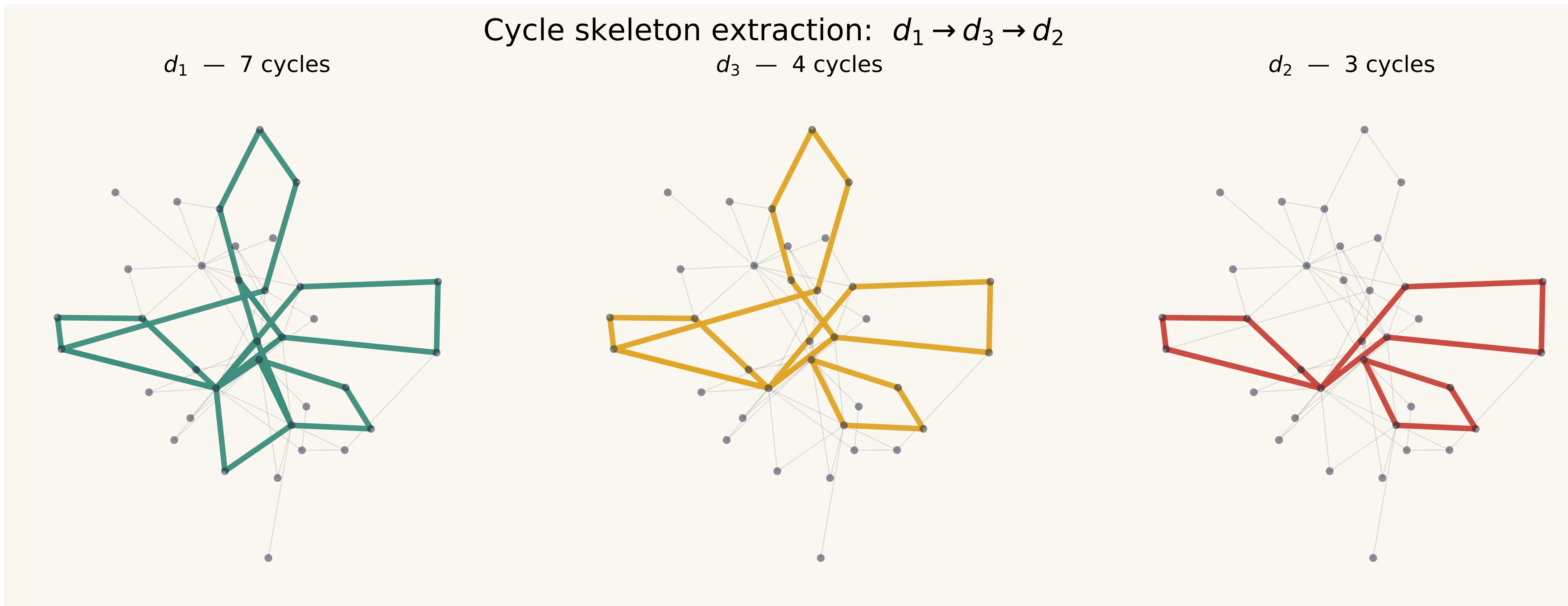
Project page

The big idea

Distance is a **tunable resolution knob** on musical structure. Sweeping $d_1 \to d_3 \to d_2$ distills a piece down to its **core cyclic skeleton**.

- Three path distances on the music graph: d_1 (min edges), d_2 (min weight, the only metric), d_3 (hybrid).
- Proven ordering $d_2 \le d_3 \le d_1$ and barcode injection $B_2 \subseteq B_3 \subseteq B_1$.
- Validated on real Korean court music — exact match to the paper's results.

Cycle skeleton extraction (Sanghyeondodeuri, geomungo)



As $d_1 \to d_3 \to d_2$, cycles vanish $\mathbf{7} \to \mathbf{4} \to \mathbf{3}$: only the robust core loops survive.

From music to a graph

Each **node** = (pitch, duration); an **edge** joins notes that occur consecutively; the **weight** is 1/(co-occurrence count) (frequent neighbors are closer). Rests are skipped.

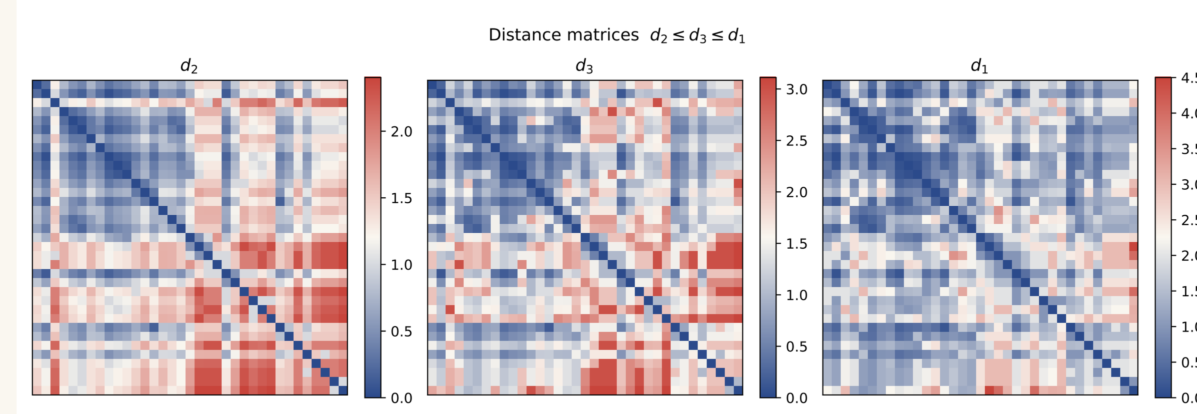
Main result: a universal ordering

For *every* pair of nodes,

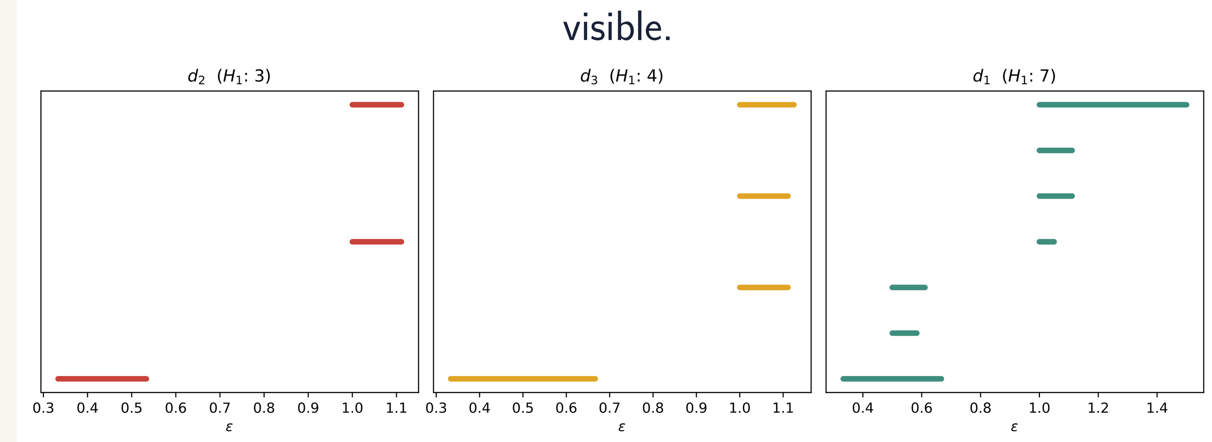
$$d_2(v,w) \le d_3(v,w) \le d_1(v,w).$$

This carries over to 1-dimensional persistent homology.

Evidence on real music



Distance matrices: d_2 (cool/smooth) $\to$ d_1 (hot) — the ordering is visible.

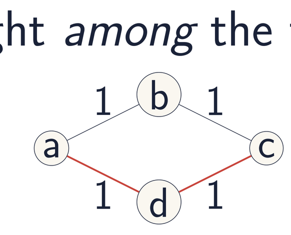


H_1 barcodes: **7** / **4** / **3** bars for $d_1/d_3/d_2$; lifespans grow, births shared.

Three path distances

For nodes v, w on the music graph $M = (V, E, W)$:

- d_1 **min-edge path** (fewest edges; weighted sum along it).
- d_2 **min-weight path** (smallest total weight) — *the only true metric.*
- d_3 **hybrid**: min weight *among* the fewest-edge paths.



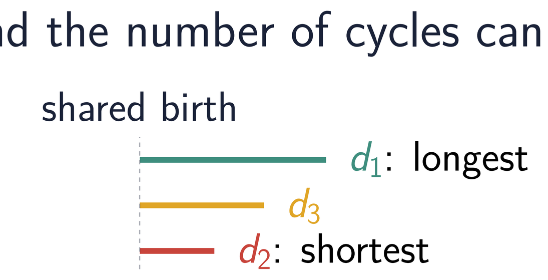
Path-based distance uses existing edges (red), not a new direct link.

Barcode injection $B_2 \subseteq B_3 \subseteq B_1$

The birth edges satisfy $B_2 \subseteq B_3 \subseteq B_1$, giving injections

$$\mathrm{bcd}_1(d_2) \hookrightarrow \mathrm{bcd}_1(d_3) \hookrightarrow \mathrm{bcd}_1(d_1).$$

Each cycle keeps its **birth**; its **lifespan grows** $d_2 \le d_3 \le d_1$, and the number of cycles can only increase.



Application: distilling the core

Ranking d_1 cycles by how far down they survive ($\to d_3 \to d_2$) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source (distance-varied generation).

Audio demos & interactive explorer on the project page.

Only d_2 is a metric

d_2 satisfies the triangle inequality; d_1 and d_3 can violate it (e.g. a heavy direct edge). Yet all three give meaningful musical structure.

Why it matters

The same piece yields *different but nested* topological readings. d_2 exposes the robust core; d_1 adds finer, peripheral loops. The changes are **systematic, not random**.

Key refs: Heo–Choi–Jung (2024) arXiv:2501.03553; Tran–Park–Jung (2023) *J. Math. Music*; Sethares–Budney (2014).

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arXiv:2506.13595